fes-then-batched, end to end (2026-11-12)

`fes-then-batched`: the two measured gains combined (2026-11-12)

Owner: “Keep working on these things.” Two levers measured today each lower the S of the algebraic index-calculus row on part of the ladder:

- `fes-f2` at 14 unknowns, through the size rule `fes-then-inherited` (`../size_rule_fold_20261112/`: degree-13 Koblitz S 101.9 against `inherited-f4`’s 138.0);
- `inherited-f4-batched` at every size (`../inherited_f4_batched_20261112/`: S 0.82 to 0.89× of `inherited-f4` on all six instances; at degree 13 its 120.2 stays above `fes-f2`’s 101.9).

The new engine `fes-then-batched` (`src/cryptanalysis/ic_framework/solvers.rs`) is the size rule with `inherited-f4-batched` above the threshold: `fes-f2` for quadratic systems of at most `fes_max_vars` = 14 unknowns, the batched engine otherwise. The threshold is the one fixed before the size-rule run. Test: `ic_framework::solvers::tests::size_rule_and_inherited_fold_match_the_reference` (extended: both branches against the exhaustive reference, word XORs no more than `fes-then-inherited`’s).

Protocol

`run_e2e.sh` under `tools/benchrec.py run`, frozen binary: `ic bench --sweep` on the six instances of the e2e ladder, the algebraic oracle with `inherited-f4`, `fes-then-inherited` and `fes-then-batched`, the pair table as reference row, 3 repeats, 60 s per call, 16 matched rho runs, one thread, local container, not isolated; no leaderboard row.

Pre-stated expectation and falsifier

Counts are deterministic for fixed seeds, so: at degree 13 `fes-then-batched` has exactly `fes-then-inherited`’s S (both run `fes-f2`); on the other five instances it has exactly the S `inherited-f4-batched` had in the batched study’s run 2 (same instances, seeds and repeats), 0.82 to 0.89× of `inherited-f4`’s. So it is the lowest algebraic row on every instance; the pair table stays below it on random 21. Falsifier: any configuration failing to recover its logarithm; `fes-then-batched` S differing from `fes-then-inherited` at degree 13, or not below `inherited-f4` on any other instance.

Results [pcc:unverified]

Run 2026-11-12 20:09 to 20:13 Z, record `benchmarks/records/20261112T2009Z-fes-then-batched-e2e.json` (measured; raw `results.tar.gz`, `results.json`), binary built at 4900647bcd from a clean tree (SHA-256 in `results/ic-sha256.txt`). Full table `RESULTS.md`, graph `results.svg/png`, `REPORT.pdf`. Every configuration recovered its planted logarithm in every repeat, and so did every matched rho run.

instance (unknowns)	rho S	pair table	inherited-f4	fes-then-inherited	fes-then-batched	ftb / inherited-f4
Koblitz 13 (14)	2.89	197.8	137.9	101.9	**101.9**	0.739
random 15 (16)	2.28	473.7	326.0	326.0	**277.5**	0.851
Koblitz 17 (18)	1.73	493.2	304.5	304.5	**249.7**	0.820
Koblitz 19 (20)	1.35	1,063.3	465.5	465.5	**413.6**	0.889
random 21 (22)	1.24	1,975.9	4,411.6	4,411.6	3,726.1	0.845
Koblitz 23 (24)	0.88	3,986.3	2,051.2	2,051.2	**1,674.3**	0.816

Mean over 3 repeats; bold where `fes-then-batched` is the lowest index-calculus row of the instance.

- **`fes-then-batched` is the lowest algebraic row on every instance** and the lowest index-calculus row on five of six (the pair table stays below it on random 21): 0.74 to 0.89× of `inherited-f4`. At degree 13 it runs `fes-f2` (S 101.9, as `fes-then-inherited`); above it runs the batched engine.
- **The Koblitz instances reproduce exactly**: S equals `fes-then-inherited`'s at degree 13 and the batched study's `inherited-f4-batched` S at degrees 17, 19 and 23 to every digit, since those rows are priced from pinned counts.
- **The pre-stated expectation was wrong in one detail**: on the random instances the absolute S differs from the batched study's (277.5 against 223.9 at random 15; `inherited-f4` moved from 262.7 to 326.0), because those rows are priced by a word-XOR ratio measured on the host in each run, not a pinned one. The ratio to `inherited-f4` in the same run is unchanged (0.851 and 0.845 in both runs). The falsifier did not fire: no failed recovery, degree 13 equal to `fes-then-inherited`, and below `inherited-f4` everywhere.
- Distance remaining to the matched rho with `fes-then-batched`: 35.3× at degree 13, 144× at Koblitz 17, 306× at 19, 1,903× at 23 (and 122× / 3,005× on the random instances).

No leaderboard row: no replay certificate or validator report (validation gate); wall time per call carries the batched engine's caveat below 22 unknowns (`../inherited_f4_batched_20261112/`, runs 1 and 3).

Sources checked

KB (`research-kb`) unavailable in this session.
`src/cryptanalysis/ic_framework/{solvers,mod}.rs` (the `measured` versus `pinned` word-XOR pricing), `research/size_rule_fold_20261112/`, `research/inherited_f4_batched_20261112/`.

Per instance: the matched counted rho (mean S over its runs) and every configuration run end to end. `S` = total group-addition equivalents / sqrt(r), mean over repeats (min–max); `S / rho` = mean s_over_rho; `ok` = every repeat recovered the planted logarithm; `hit` = decomposition hit rate; `tried` = targets tried; `solver` = engine, priced_by, mean calls, budget hits, share of the row's GAE, wall per call.

instance (r)	configuration	S mean (min–max)	S / rho
`icv1-f2m15-tm249-9e24ae34` (r = 5503)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	2.28	1.00
	`binary-subspace[dimension=8] + mitm[m=2] + – + walk`	473.73 (473.16–474.65)	207.79
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + inherited-f4 + walk`	326.04 (233.89–454.68)	143.01
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + fes-then-inherited + walk`	326.04 (233.89–454.68)	143.01
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + fes-then-batched + walk`	277.51 (200.58–386.85)	121.72
`icv1-f2m21-tm1325-`	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by`	1.24	1.00

instance (r)	configuration	S mean (min–max)	S / rho
b6925563 (r = 1049239)	doubling, distinguished points, stride starts, 16 runs		
	binary-subspace[dimension=11] + mitm[m=2] + – + walk	1975.85 (1975.82–1975.91)	1597.48
	binary-subspace[dimension=11] + descent-algebraic[m=2] + inherited-f4 + walk	4411.63 (4238.66–4648.78)	3566.82
	binary-subspace[dimension=11] + descent-algebraic[m=2] + fes-then-inherited + walk	4411.63 (4238.66–4648.78)	3566.82
	binary-subspace[dimension=11] + descent-algebraic[m=2] + fes-then-batched + walk	3726.09 (3581.30–3925.61)	3012.56
icv1-f2m13-t181-515ee569 (r = 2003)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	2.89	1.00
	binary-subspace[dimension=7] + mitm[m=2] + – + walk	197.78 (197.66–198.01)	68.38
	binary-subspace[dimension=7] + descent-algebraic[m=2] + inherited-f4 + walk	137.95 (117.79–167.62)	47.70
	binary-subspace[dimension=7] + descent-algebraic[m=2] + fes-then-inherited + walk	101.91 (83.71–116.31)	35.24

instance (r)	configuration	S mean (min–max)	S / rho
	`binary-subspace[dimension=7] + descent-algebraic[m=2] + fes-then-batched + walk`	101.91 (83.71–116.31)	35.24
`icv1-f2m17-tm101-00378d4e` (r = 65587)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	1.73	1.00
	`binary-subspace[dimension=9] + mitm[m=2] + – + walk`	493.17 (492.55–493.54)	285.87
	`binary-subspace[dimension=9] + descent-algebraic[m=2] + inherited-f4 + walk`	304.50 (102.55–415.33)	176.51
	`binary-subspace[dimension=9] + descent-algebraic[m=2] + fes-then-inherited + walk`	304.50 (102.55–415.33)	176.51
	`binary-subspace[dimension=9] + descent-algebraic[m=2] + fes-then-batched + walk`	249.66 (84.46–340.52)	144.72
`icv1-f2m19-tm797-9c54981b` (r = 262543)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	1.35	1.00

instance (r)	configuration	S mean (min–max)	S / rho
	`binary-subspace[dimension=10] + mitm[m=2] + – + walk`	1063.26 (1062.78–1063.52)	785.16
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + inherited-f4 + walk`	465.45 (227.52–594.50)	343.71
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + fes-then-inherited + walk`	465.45 (227.52–594.50)	343.71
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + fes-then-batched + walk`	413.64 (202.26–527.86)	305.45
`icv1-f2m23-tm5197-1f85e9e1` (r = 4196903)	**matched rho** `signed-Frobenius r-adding walk, distinguished points (koblitz_signed_frobenius_rho_reference)`, 16 runs	0.88	1.00
	`binary-subspace[dimension=12] + mitm[m=2] + – + walk`	3986.33 (3986.00–3986.59)	4542.18
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + inherited-f4 + walk`	2051.17 (1498.17–2448.60)	2337.18
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + fes-then-inherited + walk`	2051.17 (1498.17–2448.60)	2337.18

instance (r)	configuration	S mean (min–max)	S / rho
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + fes-then-batched + walk`	1674.33 (1223.11–1998.99)	1907.79